Aman Verma

SOFTWARE ENGINEER INTERN - ARANGO COOKBOOKS

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WORK EXPERIENCE

Software Development Engineer Intern Jan 2026 - Jun 2026

LexoLegal *Remote*

- Architected async REST APIs in **FastAPI**, **SQLAlchemy**, and **PostgreSQL** (SQL) behind an **Nginx** proxy on **Linux**, containerized via **Docker** with automated **Git** CI/CD pipelines.
- Built and deployed 'Munshi', an autonomous legal assistant in **Pydantic AI** and **FastAPI**, resolving multi-turn case inquiries with **sub-2s latency** via tool-calling loops.
- Prevented race conditions from rapid message bursts by designing **Celery** workers with atomic **Redis** distributed locks and non-blocking 10s countdown requeues (90-retry ceiling).
- Integrated **Model Context Protocol (MCP)** clients with **Pydantic AI**, isolating verified user identity inside AgentDeps rather than LLM tool parameters to eliminate prompt injection risks.
- Automated judicial data ingestion across 25 High Courts and 743 District Courts via **Celery** and **Bash** on **Linux**, utilizing **Selectolax** DOM extractors and **JSON** payloads for batch **PostgreSQL** updates.

PROJECTS

ncert-rag-evals | *Python, Jupyter Notebooks, uv, SQLite FTS5, ChromaDB, LlamaIndex, Pytest* GitHub

- Constructed an empirical retrieval benchmark across 17 STEM textbooks in **Python** and **Jupyter Notebooks**, implementing custom hybrid search in **SQLite FTS5** and **ChromaDB** instead of high-level **LlamaIndex** abstractions to profile 8 retrieval strategies on raw CPU latency.
- Proved that fusing BM25 with dense embeddings via Reciprocal Rank Fusion (RRF) achieved **86.9% Recall@5 in 8.9ms on CPU**, matching LLM query expansion while cutting latency by **99.8%**.
- Engineered a document parser indexing 4,429 section-aware chunks into **SQLite FTS5** and **ChromaDB**; validated statistical significance via paired McNemar tests (**p = 0.0021**) across 92 unit tests in **Pytest**.

doubtless | *Pydantic AI, FastAPI, Celery, Redis, SQLite, OpenTelemetry, Logfire* GitHub

- Developed an interactive lecture doubt-solver in **Pydantic AI** and **FastAPI**, fusing blackboard video frames with 45s playback dialogue windows to stream real-time solutions for visual queries via SSE.
- Designed an asynchronous 4-stage media ingestion pipeline (PipelineRunner) in **Celery** and **Redis** that transcodes HLS streams, slices 16kHz mono audio (<15MB/hr), and transcribes in <10s via Groq Whisper-Large-V3.
- Implemented the async data layer with thread-safe **SQLite** WAL repositories and **Redis** caching; instrumented distributed tracing across agent runs via **OpenTelemetry** and **Logfire** (151 tests).

vectorless-rag | *Python, Jupyter Notebooks, Matplotlib, Pydantic AI, Ollama, ChromaDB, PyMuPDF* GitHub

- Evaluated dense vector search against LLM-based page indexing on benchmark documents using **Pydantic AI**, configuring **Ollama** as a local testbed alongside cloud LLMs across 18 benchmark queries.
- Benchmarked retrieval trade-offs in **Jupyter Notebooks** using **Matplotlib**: analyzed query failure cases and plotted the **2.7x token cost** and **2x latency** tradeoff (while verifying **100% retrieval hit rate** and **0.85 citation precision**).
- Created a zero-framework document ingestion engine in **Python** using **PyMuPDF** and **ChromaDB**, streaming page-aligned windows with bounded memory (**<200MB**) to eliminate context truncation and cut extraction latency by **40%**.

TECHNICAL SKILLS

Languages & Core: Python, SQL, Bash, JSON, Linux

AI & Vector Retrieval: Pydantic AI, ChromaDB, Model Context Protocol (MCP), Ollama, LlamaIndex

Backend & Distributed Infrastructure: FastAPI, Celery, Redis, PostgreSQL, SQLite, Nginx, Docker

Evals, Benchmarking & Tooling: Jupyter Notebooks, Matplotlib, Pytest, OpenTelemetry, Logfire, PyMuPDF, Selectolax, Git

LEADERSHIP & ACTIVITIES

- Delivered 3 technical sessions as a speaker on 'Generative AI Unlocked: LLM Basics' and 'Level Up Your Career with AI', training 150+ college students on foundational LLM concepts and modern AI developer tools.
- Organized Smart India Hackathon internal rounds across 2 editions; coordinated problem statement curation, mentoring, and judging logistics for 20+ teams and 120+ participants.

EDUCATION

Veer Bahadur Singh Purvanchal University 2023 - 2027

Bachelor of Technology in Computer Science & Engineering *Jaunpur, India*